

Code

FrontEnd Code :

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>SmartElectro AI</title>
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
font-family: 'Segoe UI', sans-serif;
}
body {
background: #0f172a;
color: white;
}

/* NAVBAR */
nav {
display: flex;
justify-content: space-between;
padding: 20px 8%;
background: rgba(0,0,0,0.4);
}
.logo {
font-weight: bold;
font-size: 20px;
color: #4f46e5;
}
nav ul {
list-style: none;
display: flex;
gap: 20px;
}

/* HERO */
.hero {
height: 80vh;
display: flex;
flex-direction: column;
justify-content: center;
```

```
align-items: center;
background: linear-gradient(135deg, #1e3a8a, #0f172a);
text-align: center;
}
.hero h1 {
font-size: 42px;
margin-bottom: 20px;
}
.hero button {
padding: 12px 25px;
border-radius: 30px;
border: none;
background: linear-gradient(135deg, #4f46e5, #9333ea);
color: white;
cursor: pointer;
}

/* PRODUCTS */
.products {
padding: 60px 8%;
text-align: center;
}
.product-grid {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
gap: 25px;
margin-top: 30px;
}
.card {
background: #1e293b;
padding: 25px;
border-radius: 15px;
transition: 0.3s;
}
.card:hover {
transform: translateY(-8px);
}

/* FOOTER */
footer {
padding: 20px;
text-align: center;
background: #0f172a;
}
```

```

/* CHAT BUBBLE */
.chat-bubble {
position: fixed;
bottom: 25px;
right: 25px;
background: linear-gradient(135deg, #4f46e5, #9333ea);
color: white;
font-size: 22px;
padding: 15px 18px;
border-radius: 50%;
cursor: pointer;
box-shadow: 0 8px 20px rgba(0,0,0,0.3);
z-index: 999;
animation: pulse 2s infinite;
transition: 0.3s;
}
.chat-bubble:hover {
transform: scale(1.1);
}
@keyframes pulse {
0% { box-shadow: 0 0 0 0 rgba(79,70,229,0.6); }
70% { box-shadow: 0 0 0 15px rgba(79,70,229,0); }
100% { box-shadow: 0 0 0 0 rgba(79,70,229,0); }
}

```

```

/* CHAT CONTAINER */
.chat-container {
position: fixed;
bottom: 90px;
right: 25px;
width: 340px;
background: white;
border-radius: 20px;
box-shadow: 0 20px 45px rgba(0,0,0,0.3);
display: flex;
flex-direction: column;
overflow: hidden;
z-index: 999;
transform: translateY(40px);
opacity: 0;
pointer-events: none;
transition: all 0.35s ease;
}

```

```
.chat-container.active {  
transform: translateY(0);  
opacity: 1;  
pointer-events: auto;  
}
```

```
/* CHAT HEADER */
```

```
.chat-header {  
background: linear-gradient(135deg, #4f46e5, #9333ea);  
color: white;  
padding: 15px;  
display: flex;  
justify-content: space-between;  
align-items: center;  
font-weight: bold;  
}  
.chat-header button {  
background: none;  
border: none;  
color: white;  
cursor: pointer;  
}
```

```
/* MESSAGES */
```

```
.chat-messages {  
padding: 15px;  
height: 250px;  
overflow-y: auto;  
background: #f3f4f6;  
}  
.bot-message,  
.user-message {  
margin-bottom: 10px;  
padding: 10px 14px;  
border-radius: 15px;  
max-width: 80%;  
}  
.bot-message {  
background: #e5e7eb;  
color: black;  
}  
.user-message {  
background: #4f46e5;  
color: white;
```

```
margin-left: auto;
}
```

```
/* INPUT */
.chat-input {
display: flex;
border-top: 1px solid #ddd;
}
```

```
.chat-input input {
flex: 1;
padding: 10px;
border: none;
outline: none;
}
```

```
.chat-input button {
background: #4f46e5;
color: white;
border: none;
padding: 0 15px;
cursor: pointer;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<nav>
```

```
<div class="logo">SmartElectro AI</div>
```

```
<ul>
```

```
<li>Home</li>
```

```
<li>Products</li>
```

```
<li>Support</li>
```

```
</ul>
```

```
</nav>
```

```
<section class="hero">
```

```
<h1>AI-Powered Electronic Store</h1>
```

```
<button onclick="toggleChat()">Chat with AI</button>
```

```
</section>
```

```
<section class="products">
```

```
<h2>Popular Categories</h2>
```

```
<div class="product-grid">
```

```
<div class="card">^_□ Smartphones</div>
```

```
<div class="card">¿□_ Laptops</div>
```

```
<div class="card">^_□ Headphones</div>
```

```
<div class="card">^_□ Smart Watches</div>
```

```
</div>
```

```

</section>
<footer>
© 2026 SmartElectro AI
</footer>
<div class="chat-messages" id="chatMessages">
<div class="bot-message">Hi 5\3` I'm your AI assistant. How can I help you?</div>
</div>
<div class="chat-input">
<input type="text" id="userInput" placeholder="Ask something..." />
<button onclick="sendMessage()">> </button>
</div>
</div>
<script>
function toggleChat() {
const chat = document.getElementById("chatBox");
chat.classList.toggle("active");
}
function sendMessage()
{
const input = document.getElementById("userInput");
const message = input.value.trim();
if (message === "") return;
const messages = document.getElementById("chatMessages");
const userDiv = document.createElement("div");
userDiv.classList.add("user-message");
userDiv.textContent = message;
messages.appendChild(userDiv);
setTimeout(() => {
const botDiv = document.createElement("div");
botDiv.classList.add("bot-message");
botDiv.textContent = "Thanks for your question! AI integration coming next •'s.,7 ";
messages.appendChild(botDiv);
messages.scrollTop = messages.scrollHeight;
}, 600);
input.value = "";
}
</script>
<script
src='//in.fw-cdn.com/32724696/1539767.js'
chat='true'>
</script>
</body>
</html>

```

Embedded Code :

```
<script
src='//in.fw-cdn.com/32724696/1539767.js'
chat='true'>
</script>
```

Backend Code :

```
from flask import Flask, request, jsonify
import psycopg2
from sentence_transformers import SentenceTransformer
app = Flask(name)
model = SentenceTransformer('all-MiniLM-L6-v2')
DB_CONFIG = {
"host": "aws-1-ap-northeast-1.pooler.supabase.com",
"database": "postgres",
"user": "postgres.yabryqxxakjzfrmmzxwn",
"password": "FinalYearProjec",
"port": "5432",
"sslmode": "require",
"connect_timeout": 10
}
def get_cursor():
conn = psycopg2.connect(**DB_CONFIG)
return conn, conn.cursor()
@app.route('/chat', methods=['POST'])
def chatbot():
try:
print("=== REQUEST RECEIVED ===")
print("Raw data:", request.data)
data = request.json
print("JSON data:", data)
user_query = data.get('query', "")
print("Query:", user_query)
if not user_query:
return jsonify({"messages": [{"message": "No query received"}]})
embedding = model.encode(user_query).tolist()
print("Embedding done")
conn = psycopg2.connect(**DB_CONFIG)
cursor = conn.cursor()
print("DB connected")
cursor.execute("""
SELECT name, brand, category
FROM products_product
```

```
ORDER BY embedding <-> %s::vector
LIMIT 3;
"""', (str(embedding),))
results = cursor.fetchall()
print("Results:", results)
cursor.close()
conn.close()
if not results:
    reply = "Sorry, no matching products found."
else:
    lines = [f"{r[0]} by {r[1]} ({r[2]})" for r in results]
    reply = "Here are some products:\n" + "\n".join(lines)
    return jsonify({"messages": [{"message": reply}]})
except Exception as e:
    print("=== ERROR ===", str(e))
    return jsonify({"messages": [{"message": "Error: " + str(e)}]})
if name == 'main':
    app.run(debug=True)
```